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Generative AI education: Building a community of practice at Mayo Clinic

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ABSTRACT

Due to the rapid evolution of generative AI, we recommend a minimalist community of practice model to promote continuous social learning in an environment that fosters discussion, inquiry and collaboration while minimising administrative burden. To illustrate the relative ease with which such a model can be established and sustained, this paper describes an interdisciplinary collaborative effort by educators from multiple departments at Mayo Clinic. While we have encountered challenges, we are heartened by the engagement of our community as well as the unexpected benefits we enjoy as the founding members and facilitators of the community.

KEYWORDS

Generative AI; health professions education; higher education; learning communities; communities of practice

1. Introduction/Background

Every so often something new comes along that changes what is possible. In November 2022, OpenAI released ChatGPT and sparked the proliferation of a new technology that raised many questions for educators:

- What are the risks of generative AI?
- How do you keep up with and apply these emerging tools?
- How do you grow confidence with tools that are changing constantly?

This opinion piece explains why we believe the answer lies in the community. It describes how our experiences illustrate that the community of practice (CoP) model is a viable option for any group in higher education that wants to take an active role in growing their confidence and competency with generative AI.

In October 2023, we launched a virtual Generative AI in Education CoP at Mayo Clinic to bring together healthcare educators to learn about generative AI by asking questions, sharing knowledge and discussing practical use cases. A CoP is a group with a common

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interest that meets periodically to share ideas and engage in a rich dialogue to problem-solve and advance the collective understanding of a topic (de Carvalho-Filho et al., 2020). Although the CoP concept has evolved over time (Li et al., 2009), its core idea of social learning in an organisation to promote knowledge-sharing is as relevant as ever. It can be challenging to learn about tools that do not come with user manuals and change constantly, particularly if they come with ethical and policy implications. A CoP is a space that brings to life evidence-based and applied learning (Brooks et al., 2024). Our CoP realises those ideals: it is practical, meaningful and community-driven, while grounded in both theory and practice. What is unique about our CoP, however, is how easy it has been to establish and maintain, contrasting other CoP models (Brooks et al., 2024).

Over the past year, we learned about establishing and sustaining a CoP responsive to the needs of our participants while supporting the digital transformation goals of our institution. We also learned that there are unexpected benefits for us as facilitators. We are excited to share our lessons learned, including how our CoP is structured, what works well, our challenges and the overall power of learning in a community.

2. Building the community

Our facilitation team brings unique perspectives from our diverse roles at Mayo Clinic, united by our collective goal to grow understanding of generative AI in education. Our monthly planning process is dynamic, adapting to ever-evolving generative AI tools, topics and guidance. Our roles emerged organically, dividing the responsibilities of preparing and promoting upcoming sessions. All four of us co-facilitate discussions, which minimises disruptions when one of us cannot attend a session.

We intentionally keep our CoP sessions informal to create a space where attendees feel comfortable asking questions, sharing insights and building connections. Each month, our CoP sessions became more sophisticated and nuanced as participants' knowledge of generative AI grew. The progression highlights how the community not only learns together but also shapes the content itself, an essential principle of CoPs (de Carvalho-Filho et al., 2020). This progression also serves as a responsive way to address the sense of never being up-to-speed when tools change constantly and are impossible to master completely (Craft, 2024).

3. Successes and challenges

As we complete the first year of our CoP, we are reflecting on our successes and challenges thus far. We are proud of the range of topics and formats we offered for our sessions (Table 1).

To keep the content fresh and engaging, we alternate topic themes from month-to-month. Our attendance grew steadily throughout the year (Figure 1), largely through word-of-mouth. Every session included new members, indicating an extension in our reach. We also see the impacts of our CoP demonstrated within our sessions when our members present use cases inspired by their learnings from previous sessions.

We frequently seek attendee feedback to provide members with a sense of shared ownership of their learning (Rainer & Matthews, 2002) and to help plan sessions. With so

Table 1. Community of practice session topic themes from the first year.

Topics	# of Sessions
‘How to’ generative AI tool demonstrations	3
Use case presentations	3
Generative AI in education podcast discussions	3
Opportunities and challenges to implementing generative AI discussions	2
Ethics of generative AI	2

We held 12 sessions in the first year. However, one session included multiple topics, which accounts for the sum of 13 in this table. Tool demonstrations examples included image generation for medical education, using ChatGPT to summarise podcast transcripts and to develop discussion questions and using Perplexity AI to locate and synthesise resources. Use case presentations demonstrated how Mayo Clinic education colleagues use generative AI in their work.

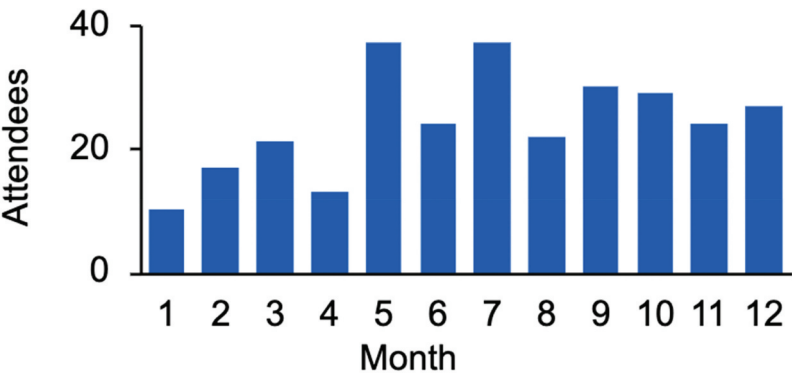


Figure 1. Number of attendees over 1 year.

many potential topics as generative AI quickly evolves, feedback from members helps us prioritise the most salient and urgent topics. Our members are eager to explore applications of generative AI within education at Mayo Clinic. We often hear from colleagues that they are excited by AI but overwhelmed; they are unsure where to begin or how to engage, particularly given organisational constraints and concerns about data privacy. Our CoP reduces some of these barriers by centring data security and our organisation’s regulations in our conversations. It is one thing to learn about a trendy new AI tool, but it is another to see its direct use in health professions education. By learning from peers about how generative AI is being applied at our institution, rather than simply learning what generative AI can theoretically do, participants are better prepared to integrate it into their work.

One early challenge we noted was attrition associated with breakout rooms. We wanted to leverage breakout rooms because they encourage deeper conversation among learners (Read et al., 2022). However, we noticed several attendees left the session when breakout rooms began. This is not unique to our CoP, as we noticed this in other virtual gatherings as well. We know breakout sessions can be anxiety-inducing and that participants may not be able to engage in breakout discussions if they are in a space where they cannot use their microphone or camera. Therefore, we planned full-group sessions so community members could participate by speaking up, using the chat tool or by observing. This flexibility is key, allowing for engaging conversations while providing a low-stress environment for attendees.

Lastly, a known challenge in building a CoP is the resources required to successfully launch and maintain the community, with some organisations using dedicated staff/departments to support CoPs (Brooks et al., 2024). However, we found that with deliberate planning, a CoP can thrive with minimal administrative burden on its facilitators by distributing the workload and leveraging existing tools within the institution. Moreover, we found unexpected benefits in our facilitator planning meetings; regularly brainstorming together has improved our sense of belonging within our organisation. Building strong relationships and being known by colleagues are important to remote workers' sense of belonging (Dery & Hafermalz, 2016), and this experience of establishing and co-facilitating a CoP around a topic that is important to us increased our connection. Facilitating the CoP creates a feeling of doing our work 'together', even when working within our separate teams. Furthermore, it helps us build relationships with fellow educators who we may not have otherwise met, allowing for additional formal collaboration and informal connection outside of the CoP sessions.

4. Lessons learned

We attribute the success of our CoP to four factors that we believe can aid others interested in establishing a CoP:

- (1) Establish a small team of volunteer leaders – We need to be attuned to the developments in generative AI and potential implications and uses in education. With four facilitators, the administrative load for staying on top of trends in generative AI, planning and facilitating sessions is manageable and enjoyable (Figure 2).
- (2) Maintain an informal approach – Keeping our effort informal keeps it simple. Each month we meet once to plan the session, complete assigned tasks and then facilitate the session.
- (3) Use existing tools – We use Microsoft Teams, email and a SharePoint site to communicate and organise. This allows us to leverage what is available without additional effort.
- (4) Complement other efforts – Our CoP is one of the several efforts within our organisation to promote generative AI education. Working together with those efforts, while setting a clear intention and audience for our initiative, has amplified their impacts.

5. Future directions

As we look to the future of our CoP at Mayo Clinic, we will build upon the foundation we established over the past year. We will be formally evaluating our CoP to directly measure the impact our sessions have had on our participants' work at Mayo Clinic using the Value-Creation Framework (Wenger-Trayner et al., 2019). Our journey thus far has shown us the power of community-driven learning, and we are committed to expanding our impact while staying true to our core principles of informally yet intentionally fostering knowledge exchange and connection.

Mayo Clinic Generative AI in Education Community of Practice Facilitator's Minimalist Framework

Three Weeks Prior to Session: Facilitator Planning Meeting (one hour)

- ☐ Debrief previous CoP session
- ☐ Discuss plans for next session, including identifying general area of focus and format (e.g., AI tool demonstrations and guided practice sessions, podcast or journal discussion, use case presentations). Identify potential guest presenters based on topics selected, if needed
- ☐ Note additional ideas for future sessions as they emerge via a shared planning document (e.g., Microsoft OneNote)

Two Weeks Prior to Session: Asynchronous Preparation

- ☐ Write CoP description
- ☐ Confirm guest presenters (as needed) and send calendar invites to presenters
- ☐ Discuss any remaining logistics via facilitator's group chat on Microsoft Teams

One Week Prior to Session: Promotion

- ☐ Post CoP session description on event intranet/SharePoint
- ☐ Email reminder with session description to all registered attendees
- ☐ Post reminder and description in AI Education Teams Channel

Immediately After Session: Wrap-Up

- ☐ Post recording on event intranet/SharePoint site, which allows anyone within Mayo Clinic to access past recordings, topics, and resources from our sessions
- ☐ Record attendance information and note suggestions and chat topics from session to inform future session ideas
- ☐ Use facilitator chat to continue discussions of potential topics prior to next planning meeting

Figure 2. Mayo Clinic generative AI in education community of practice facilitator's minimalist framework.

We encourage educators, faculty developers and leaders to consider how a minimalist CoP model may work for your organisation. As we continue to learn about generative AI, and as the technology evolves, the power of learning together can transform how we approach our work while promoting both a sense of community and innovation in higher education.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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